

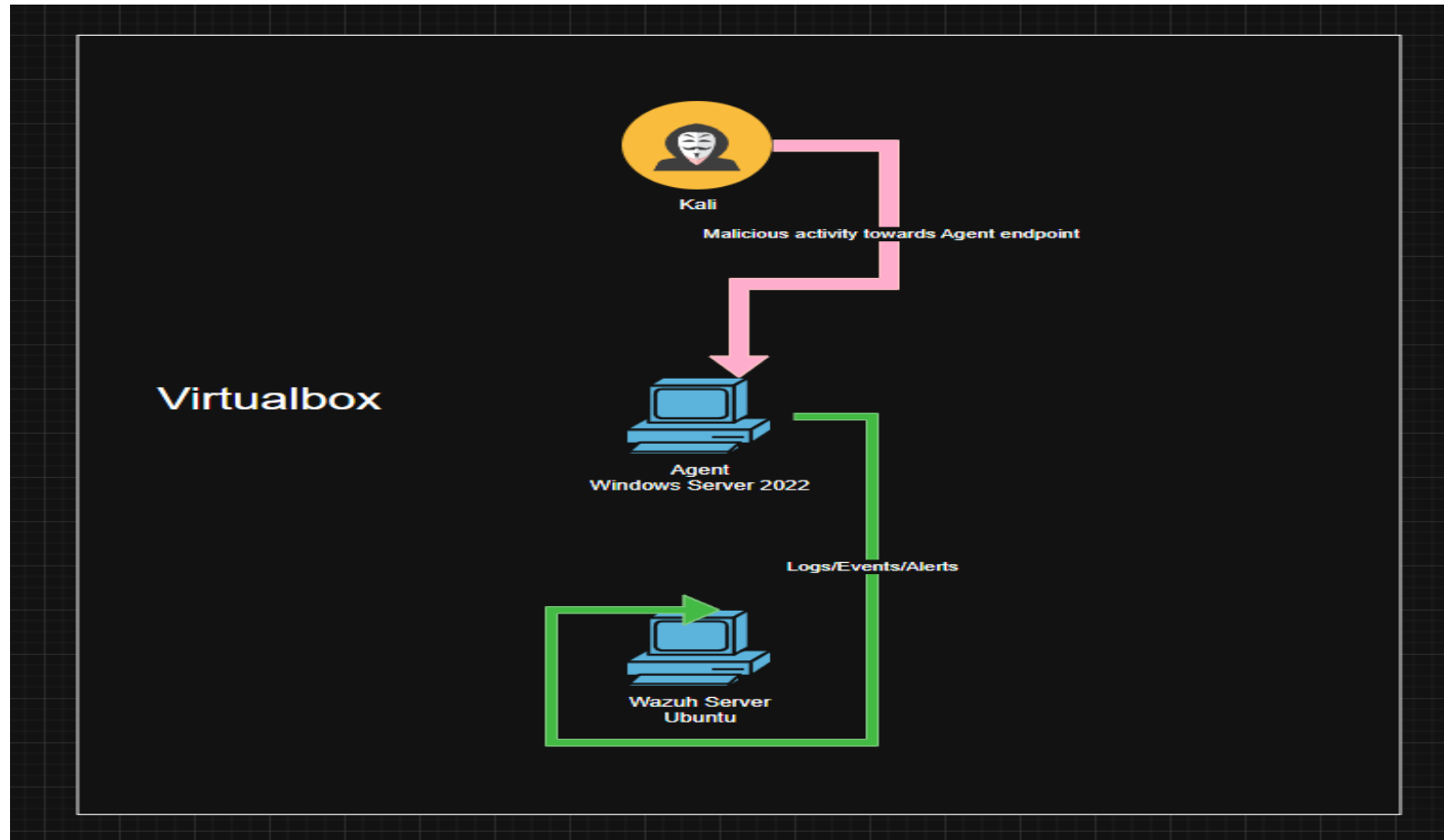
WAZUH SIEM IMPLEMENTATION & TESTING

Martin Glas 2026

Machines used

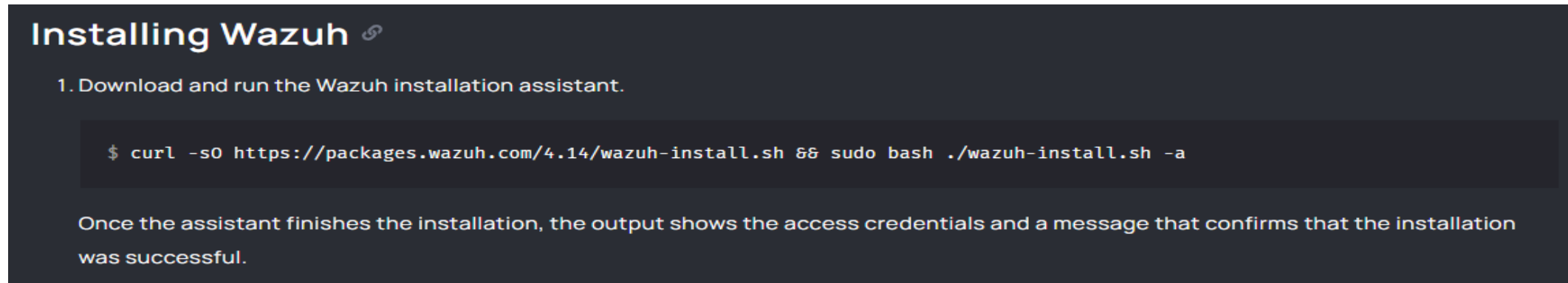
	Kali ➡ Running
	wServer2022 ➡ Running
	UbuntuWazuh ➡ Running

Infrastructure



Installation

- Wazuh installed to Ubuntu 26.04 via Quickstart recommendations



- Wazuh Agent and Sysmon installed to Windows Server 2022

Rules and File Integrity

- Configured rules to "local_rules.xml" to detect possible OpenSSH brute force

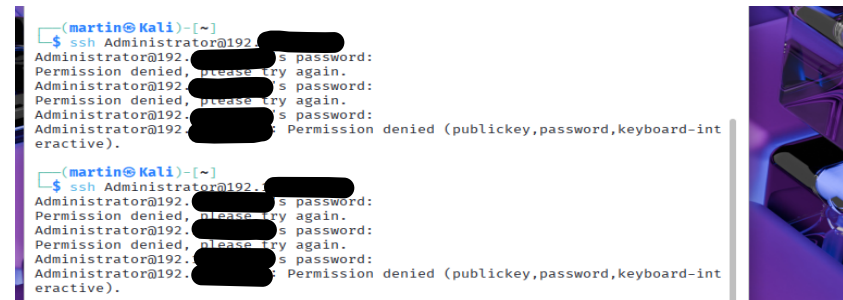
```
28 <group name="authentication_failed">
29
30 <!-- Base: failed login via SSH -->
31 <rule id="110200" level="5">
32 <if_sid>60122</if_sid>
33 <description>Windows SSH login failed</description>
34 <group>authentication_failed</group>
35 </rule>
36
37 <!-- Correlation: brute force -->
38 <rule id="110201" level="10" frequency="3" timeframe="60">
39 <if_matched_sid>110200</if_matched_sid>
40 <description>SSH brute force detected on Windows OpenSSH</description>
41 <group>authentication_failed</group>
42 </rule>
43
44 </group>
```

- Added realtime file integrity monitoring to path C:\Users\Agent

```
<directories realtime="yes">C:\Users\agent</directories>
```

OpenSSH Alerts

- Simulated brute force towards Wazuh Agent via OpenSSH to see if our rule works as intended.



```
(martin@Kali)-[~]  
$ ssh Administrator@192.168.1.102  
Administrator@192.168.1.102's password:  
Permission denied, please try again.  
Administrator@192.168.1.102's password:  
Permission denied, please try again.  
Administrator@192.168.1.102's password:  
Administrator@192.168.1.102's password:  
Permission denied (publickey,password,keyboard-interactive).  
  
(martin@Kali)-[~]  
$ ssh Administrator@192.168.1.102  
Administrator@192.168.1.102's password:  
Permission denied, please try again.  
Administrator@192.168.1.102's password:  
Permission denied, please try again.  
Administrator@192.168.1.102's password:  
Administrator@192.168.1.102's password:  
Permission denied (publickey,password,keyboard-interactive).
```



- Alerts clearly show indicators of a possible brute-force attack.

Export Formatted Reset view 748 available fields Columns Density 1 fields sorted Full screen						
timestamp	agent.name	rule.description	rule.level	rule.id		
Jun 19, 2026 @ 15:36:09.5...	winServer2022	SSH brute force detected on Windows OpenSSH	10	110201		
Jun 19, 2026 @ 15:36:09.5...	winServer2022	Windows SSH login failed	5	110200		
Jun 19, 2026 @ 15:36:08.6...	winServer2022	Windows SSH login failed	5	110200		
Jun 19, 2026 @ 15:34:54.7...	winServer2022	SSH brute force detected on Windows OpenSSH	10	110201		
Jun 19, 2026 @ 15:34:54.7...	winServer2022	SSH brute force detected on Windows OpenSSH	10	110201		
Jun 19, 2026 @ 15:34:53.8...	winServer2022	Windows SSH login failed	5	110200		
Jun 19, 2026 @ 15:30:26.7...	winServer2022	Windows SSH login failed	5	110200		
Jun 19, 2026 @ 15:30:25.8...	winServer2022	Windows SSH login failed	5	110200		
Jun 19, 2026 @ 15:30:25.8...	winServer2022	Windows SSH login failed	5	110200		
Jun 19, 2026 @ 15:22:09.5...	winServer2022	Windows SSH login failed	5	110200		
Jun 19, 2026 @ 15:22:09.5...	winServer2022	Windows SSH login failed	5	110200		
Jun 19, 2026 @ 15:22:08.6...	winServer2022	Windows SSH login failed	5	110200		
Jun 19, 2026 @ 15:19:39.2...	winServer2022	Windows SSH login failed	5	110200		
Jun 19, 2026 @ 15:19:39.2...	winServer2022	Windows SSH login failed	5	110200		
Jun 19, 2026 @ 15:19:37.9...	winServer2022	Windows SSH login failed	5	110200		

File Integrity

- Created new text file to confirm that our config works
- As file integrity logs show, it clearly works and we can see valuable data on what has been modified on the target

	Jun 19, 2026 @ 10:07:08.9...	winServer2022	c:\users\agent\desktop\shadysoftware\shadyscript.txt	modified	Integrity checksum changed.	7	550
	Jun 19, 2026 @ 10:07:01.5...	winServer2022	c:\users\agent\desktop\shadysoftware\shadyscript.txt	added	File added to the system.	5	554
	Jun 19, 2026 @ 10:07:01.4...	winServer2022	c:\users\agent\desktop\shadysoftware\new text docum...	deleted	File deleted.	7	553
	Jun 19, 2026 @ 10:06:58.0...	winServer2022	c:\users\agent\desktop\shadysoftware\new text docum...	added	File added to the system.	5	554

SUMMARY

- Implementating Wazuh Server and Agents is particularly complicated, but defining rules and monitoring the logs to spot possible attacks are.
- I got confidence on understanding Wazuh as SIEM, but rules and what to monitor besides simple self made simulations are still difficult.
- I consider the lab as success, and it still has million ways to improve.

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